

Signal Combination

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Introduction

In this report, we document three methodologies for combining alpha signals used by the BYU Silver Fund Quant team: Fama-Macbeth, Mean-Variance Efficient (MVE), and Bayesian Normal-Normal. In this report, we refer to “signal weights” as the weights assigned to each signal in the portfolio. These weights are non-negative and sum to one. We document our approach with three classic Silver Fund signals: reversal, momentum, and betting against beta (BAB).

Code

The code for this research is available on [GitHub](#).

Fama-Macbeth

The Fama-Macbeth method is inspired by the Fama-Macbeth two-stage regression. In stage one, the coefficients of each alpha signal for each security are estimated using next-period returns. These time series regressions are regressed on a rolling 5-year window for each security. This would amount to about 3,000 regressions per day for the Silver Fund universe. We will call the coefficients of this regression betas. The betas tell us how much additional return we should expect in the next period per unit change in the alpha signal for a specific security. Note that the alpha signals are computed using the traditional Grinold and Kahn formulation. This means that they are return forecasts.

$$r_{i,t+1} = \beta_0 + \sum \beta_{i,k} \alpha_{k,t} + \epsilon_{i,t}$$

During the second stage, we estimate the risk premium for each signal using cross-sectional regressions — one for each day. We regress next period returns on the betas from the previous regression. The coefficients of this new regression model will be called gammas. The gammas tell us how much return we should expect in the next period per unit change in beta across all assets. It is the cross-sectional approach that lets us then generalize the risk premiums to weights.

$$r_{i,t+1} = \gamma_0 + \sum \gamma_{k,t} \beta_{i,k} + \eta_{i,t}$$

The gammas from the second regression are taken each day and divided by the sum of their absolute values. This creates weights constrained to unit leverage, but they don't sum to 1 (not fully invested) and can be non-negative (not long-only). We further apply the softmax transformation to ensure the weights sum to one and are non-negative. Finally, we apply a one-year exponential weighted moving average (EWMA) to smooth the signal weights over time. See Figures 1, 2, and 3 for the gammas, normalized gammas, and weights.

Mean-Variance Efficient

In this method, we treat each alpha signal as a separate synthetic asset. We compute each alpha signal's returns by running an MVE backtest where the portfolio's weights sum to zero,

and the portfolio beta is constrained to one. For each alpha signal, we target an active risk of 5%. We take the time-series returns for each alpha signal and then use the analytical MVE formulation to find the optimal signal weights on a five-year rolling window. This involves computing the mean return for each signal and the covariance matrix among them, once per rolling-window iteration.

$$w_s^* = \Sigma^{-1} \mu$$

Since there are initially no constraints on these weights, we apply the same normalization, softmax, and EWMA transformations as described in the Fama-Macbeth method. See Figures 4, 5, and 6 for the raw, normalized, and final weights of this method.

Bayesian Normal-Normal

The Bayesian method follows the same procedure as the MVE approach, but with a shrinkage component that pulls the estimated variables toward the prior. We model the joint distribution of returns for each alpha signal using the Bayesian Normal-Normal conjugate prior. For more information on conjugate priors, see [Wikipedia](https://en.wikipedia.org/wiki/Conjugate_prior). Using a conjugate prior gives us a closed-form solution for the posterior distribution of returns, allowing us to regularize our estimates with our priors.

$$p(\mu \mid \mathcal{D}, \Sigma) \propto p(\mathcal{D} \mid \mu) p(\mu)$$

For this report, we model the prior as a multivariate normal distribution with a mu vector of all 1/3—this is equivalent to equal-weighting the signals—and a covariance matrix equal to the identity matrix times 1%—this is equivalent to 1% daily variance.

$$p(\mu) = \mathcal{N}(\mu \mid \check{m}, \check{V}) \quad \check{m} = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right] \quad \check{V} = \frac{1}{100} I$$

The likelihood is modeled as a multivariate normal distribution where $\bar{y}$ is the sample mean return of each signal over a five-year rolling window, and Sigma is the sample covariance matrix over the same period. Note that these two parameters are the same parameters used to find the optimal weights for the MVE method.

$$p(\mathcal{D} \mid \mu) = \mathcal{N}(\bar{y} \mid \mu, \frac{1}{N} \Sigma)$$

Finally, the posterior distribution is obtained by multiplying the likelihood and the prior probability density functions. We leave the derivation to the reader. This gives us a posterior estimate of the mean vector and covariance matrix. A key containing all of the notation is listed in Figure 11.

$$p(\mu \mid \mathcal{D}, \Sigma) = \mathcal{N}(\mu \mid \hat{m}, \hat{V}) \quad \hat{V}^{-1} = N_0 \check{V}^{-1} + N \Sigma^{-1} \quad \hat{m} = \hat{V}(\Sigma^{-1}(N \bar{y}) + \check{V}^{-1}(N_0 \check{m}))$$

We take the posterior mean ($\hat{m}$) and the posterior covariance matrix ($\hat{V}$) and compute the MVE weights using the formula for optimal portfolio weights given in the previous section. Note that the mean vector is the data-weighted combination of the likelihood and posterior

mean. We add an additional parameter, N_0 , which denotes the number of pseudo-observations for the prior. We can increase N_0 to strengthen our priors. In this report, we set the number of pseudo-priors to equal the number of observations in the rolling window. We note that this is a very aggressive prior, but we do so to illustrate the regularization property. We take the prior-adjusted weights and apply the normalization, EWMA, and softmax transformations as in the previous methods. See Figures 7, 8, and 9 for the raw weights, normalized weights, and final weights.

Results

Overall, we find agreement among the three methods regarding relative signal strength. Reversal is the clear winner among the three, and momentum and BAB are relatively similar. We also find that the Bayesian approach serves as a regularized version of the MVE approach. Note that none of the alpha signals accounts for transaction costs. This leads to reversal having a somewhat inflated weight in the Fama-Macbeth and MVE methods. This is where the Bayesian method shines, since it prevents us from assigning very large weights to a single signal.

In addition to analyzing the signal weights over time for each method, we compute the returns to each weighting method. We also compute the equal-weight baseline.

Method	Equal	Fama-Macbeth	MVE	Bayesian
Sharpe Ratio	1.48	1.67	1.68	1.59

We are excited to observe that every method outperforms the baseline. While the Bayesian method performs marginally worse than the other two methods, we believe it is still the superior strategy because it allows us greater flexibility in establishing our beliefs. We include the cumulative performance of the Bayesian strategy in Figure 10.

Conclusion

We recommend implementing the Bayesian Normal-Normal method into the current Silver Fund trading system. This can be done by adding signal weights and a combined alpha flow to the quant team data pipelines. The Bayesian Normal-Normal method is our champion weighting strategy because it retains the same properties as the MVE method but allows us to flexibly apply priors to regularize the weights, pulling them closer to our prior beliefs while also accounting for observed returns.

Appendix
Figure 1: Fama-Macbeth Gammas

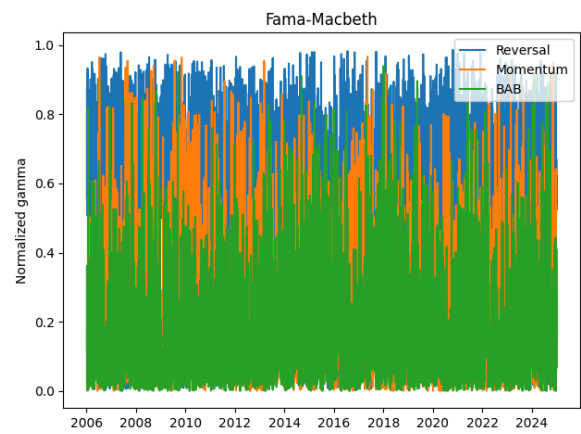


Figure 2: Fama Macbeth Normalized Weights

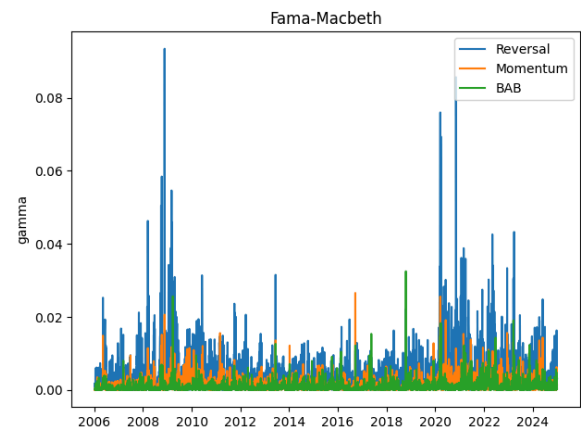


Figure 3: Fama Macbeth Signal Weights

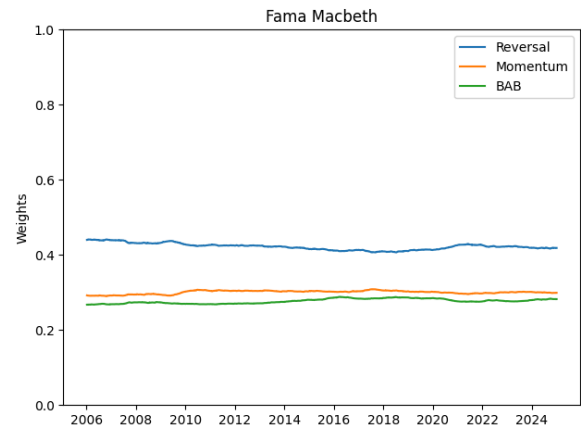


Figure 4: MVE Raw Weights

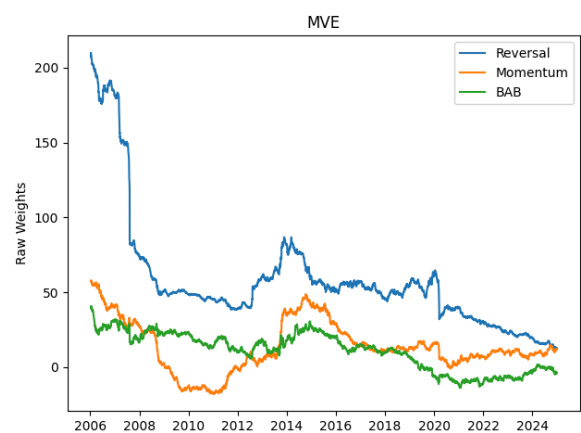


Figure 5: MVE Normalized Weights

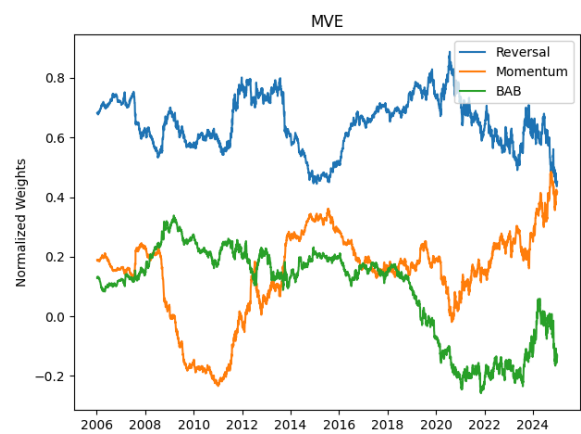


Figure 5: MVE Signal Weights

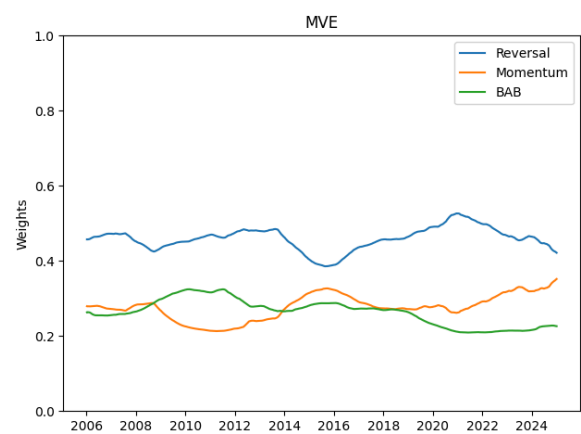


Figure 7: Bayesian Raw Weights

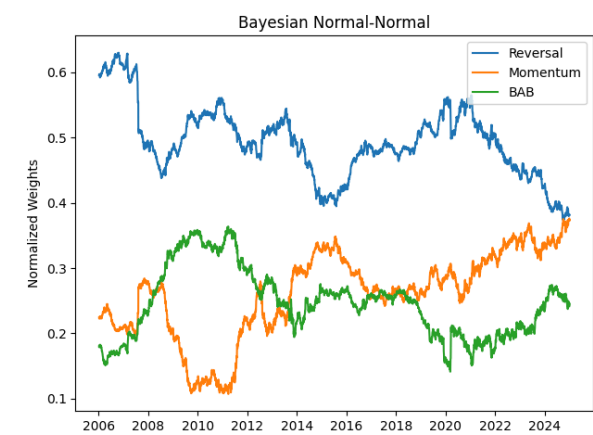


Figure 8: Bayesian Normalized Weights

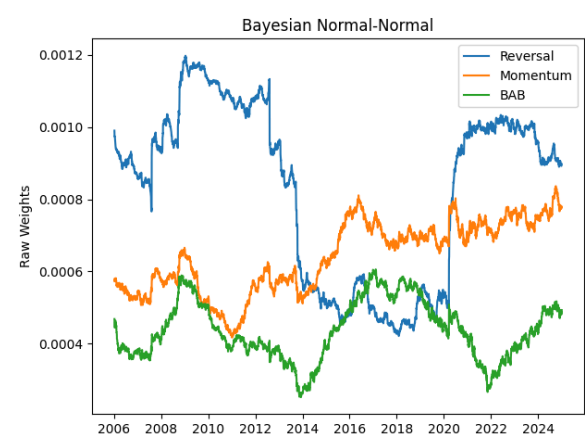


Figure 9: Bayesian Signal Weights

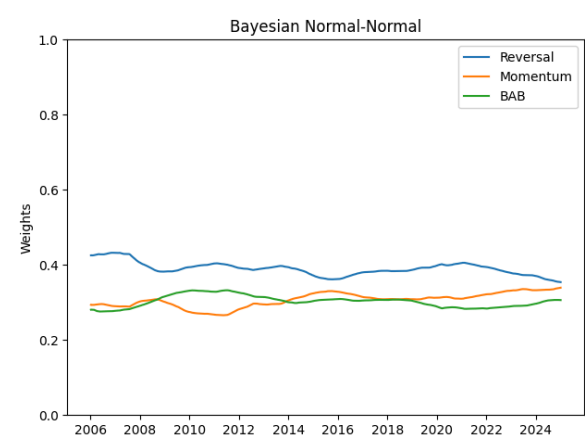


Figure 10: Bayesian Cumulative Performance

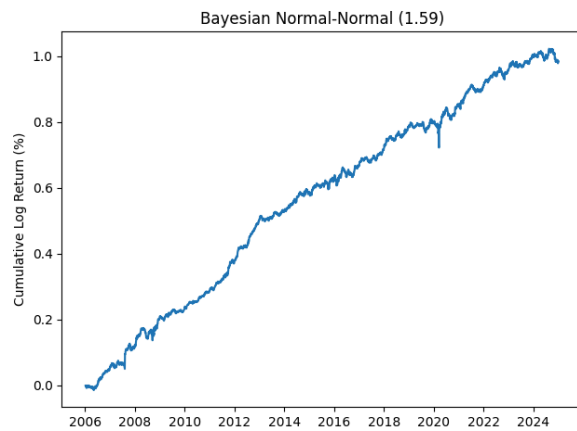


Figure 11: Symbol Key for Bayesian Method

μ : Unknown mean vector

$\hat{m}$: Posterior mean vector

$\hat{V}$: Posterior covariance matrix

$\bar{y}$: Sample mean vector

Σ : Sample covariance matrix

N : Number of sample observations

$\check{m}$: Prior mean

$\check{V}$: Prior covariance matrix

N_0 : Number of pseudo prior observations